

- 2 (a) By referring to both kinetic energy and potential energy, explain what is meant by the internal energy of an ideal gas.

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..... [2]

- (b) A fixed mass of an ideal gas at a temperature of 20°C is sealed in a cylinder by a piston, as shown in Fig. 2.1.

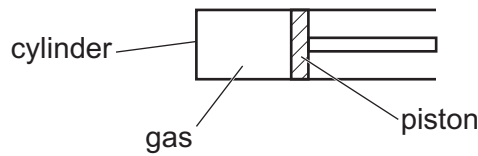


Fig. 2.1

The initial volume of the gas is $1.24 \times 10^{-4} \text{ m}^3$.

Thermal energy is supplied to the gas and its volume increases by $5.20 \times 10^{-5} \text{ m}^3$.

- (i) The piston is freely moving so that the gas is always at atmospheric pressure.

Atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$.

Calculate the work done by the gas.

work done by gas = J [2]

- (ii) Calculate the final thermodynamic temperature T of the gas.

$$T = \dots\dots\dots \text{K} \quad [2]$$

- (iii) The mass of the gas is 16 g. For this expansion, there is a net transfer of 960 J of thermal energy to the gas.

Calculate the specific heat capacity c of the gas at this pressure.

$$c = \dots\dots\dots \text{J kg}^{-1} \text{K}^{-1} \quad [2]$$

- (c) The gas in (b) is allowed to return to its starting temperature. The piston is now fixed in position.

Thermal energy is supplied to increase the temperature to the same final temperature as in (b).

Use the first law of thermodynamics to suggest and explain how the specific heat capacity of the gas for this situation compares with the value in (b)(iii).

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 [3]

[Total: 11]